TRACK 03 · DEEPFAKE DETECTION FOR FINANCIAL COMMUNICATIONS

Satyapan

सत्यापन · “verification” in Hindi

Payment verification that a deepfake cannot pass.

ONE-SENTENCE SOLUTION

Satyapan makes a deepfaked CFO or vendor **unable to move money**: every payment is checked against the company’s own vendor master, the real executive’s voice and approvals carry a cryptographic **Satyapan Seal**, and any doubt hands verification to a person and channel the attacker does not control.

> **“A deepfake can copy a CFO’s voice, never his key.”**
> *Awaaz nakli ho sakti hai, mohar nahi.* (A voice can be faked; the seal cannot.)

REVIEWER MAP · RUBRIC CRITERION → PAGE

Problem fit and user (20)	**2, 3**	Safety, privacy, responsible AI (15)	**9, A3**
Architecture and detection (20)	**4, 5, 7, A1-A3**	Adoption and scale (10)	**10, A1**
Value and differentiation (15)	**3, 5, 6**	Evidence ledger, every number	**X1**
Feasibility and 48 h plan (15)	**8, 10**	AI, data and asset disclosures	**X5**

CHALLENGE STATEMENT WE ANSWER

“Help people and institutions verify financial communications, detect manipulation, and create an actionable path for review, reporting, or response.”

Verify	Seal proves the real executive; master, the real account
Detect	provenance, forensic, behavioural signals, *raise-only*
Act	who verifies, checker hold, 1930 / NCRP / bank pack

TEAM · DELHI TECHNOLOGICAL UNIVERSITY (DTU)

Shivendra Narayan Tripathi	Lead · policy engine, Seal, capture
Saarika [surname]	Ask detection, side panel, voice
Kartikey [surname]	Email, invoice, voice checks, report
Saksham [surname]	Console, vault, OOB-Confirm, EN/HI

WHAT ALREADY RUNS (PROTOTYPE, 8 OCT 2026)

- Decision engine with 7 property-tested safety invariants
- Satyapan Seal module, API, demo pages, side-panel chip
- Ask detection EN / Hinglish / Devanagari; EN + HI text
- 511 automated tests pass; gaps listed on pages 8 and X4

Synthetic and public data only. Every number is traced in X1.

The deepfake “boss scam” has reached India’s finance desks

IT IS HAPPENING IN INDIA, NOW

JULY 2026 · ₹10.70 CRORE · LISTED-COMPANY SUBSIDIARY [R1]

Starmangalsutra Pvt Ltd, a subsidiary of Sky Gold & Diamonds, disclosed that an employee sent funds to unknown accounts on instructions from an **impersonated director**. The company cited deepfake techniques, including profile cloning, and a compromised company phone and laptop. The director had given no such instruction.

17 JULY 2026 · SEBI CAUTION TO LISTED COMPANIES, PR 40/2026 [R2]

“Boss scam”: CEO/MD impersonation using **AI deepfakes and voice cloning on video calls**, hijacked WhatsApp Web sessions, the attacker’s number saved under the CEO’s name, and “don’t tell anyone, it is price-sensitive”. Then a mule account. SEBI’s advice: **verify directly with the senior official, never pay on a message alone, report on 1930.**

2024 · ARUP, HONG KONG · HK$200M (ABOUT US$25.6M) [R3]

An employee made 15 transfers after a video call with a deepfaked CFO and colleagues. The person who should have verified was the person being deceived.

> Scale: Indian citizens lost **₹22,845.73 crore** to cyber fraud in 2024 (Lok Sabha reply, 22 Jul 2025; mostly consumer fraud) [R4]. Corporate losses come on top, and most are never disclosed.

PRIMARY USER

Priya, accounts-payable maker at a ₹80-crore Pune manufacturer (illustrative). Pays vendors by NEFT / RTGS / UPI. Checker: **Arjun**. CFO: **Rajesh**.

HARMFUL MOMENT

Priya is about to pay a **new or changed bank account**, in a hurry and in secret, because “the CFO” or “the vendor” said so.

EXACT THREAT PATTERN

1 Impersonate	2 Ask	3 Pressure	4 Pay	5 Drain
deepfake voice or video, cloned profile	new account, “keep it confidential”, “skip Arjun”	the maker is also the one who would verify	NEFT / RTGS / UPI to a mule account	moved on in hours; needs the UTR fast

HARM

Loss of lakhs to crores in minutes. Recovery depends on reporting the transaction reference (UTR) within hours. Listed firms must disclose; Sky Gold’s shares fell on the news [R1].

SHARPLY BOUNDED PROBLEM

In scope: payment instructions reaching an SME or mid-market AP team by email, WhatsApp text or voice note, and browser meetings (Google Meet, Teams on the web).

Out of scope (MVP): consumer scams, phone-call interception, native Teams / Zoom desktop apps (upload the recording instead).

[R1] Sky Gold disclosure to BSE, 16 Jul 2026 (The Economic Times) · [R2] SEBI PR 40/2026 · [R3] Fortune, May 2024 · [R4] Lok Sabha, 22 Jul 2025 · links: X6

Value and differentiation (15) · Problem fit (20)

Before and after: verification moves away from the person being deceived

	1 · INSTRUCTION ARRIVES	2 · CHECK	3 · VERIFY	4 · APPROVE AND PAY	5 · IF IT WAS FRAUD
TODAY Priya decides alone	"Rajesh" on Meet or WhatsApp: **urgent, confidential**, pay this new escrow account today.	Priya judges by **ear and face**. A detector, if any, gives a % that drifts with each new generator.	She calls the number in the message or her phone (attackers rename contacts [R2]). **The verifier is the person being deceived.**	"Approved" arrives by email or chat, which can also be faked. The amount or account can change afterwards.	Hunts for the UTR, the bank's fraud email and the NCRP form while the money moves on.
WITH SATYAPAN	Payment entered. **Floor from the vendor master:** new account ⇒ **VERIFY** whatever the message says.	Side panel: **"Claims to be Rajesh: NOT sealed"**. Asks: new account, secrecy, skip approval ⇒ **HOLD**	Only the **checker** may clear a HOLD, after his own callback on the number on file. OOB-Confirm reaches the **real** Rajesh.	Real Rajesh's passkey **signs the exact amount and account** (Intent Seal). Any change breaks the signature.	One click: **1930 script with the UTR on line 1**, bank fraud-desk email, NCRP draft, hashed evidence. A human submits.

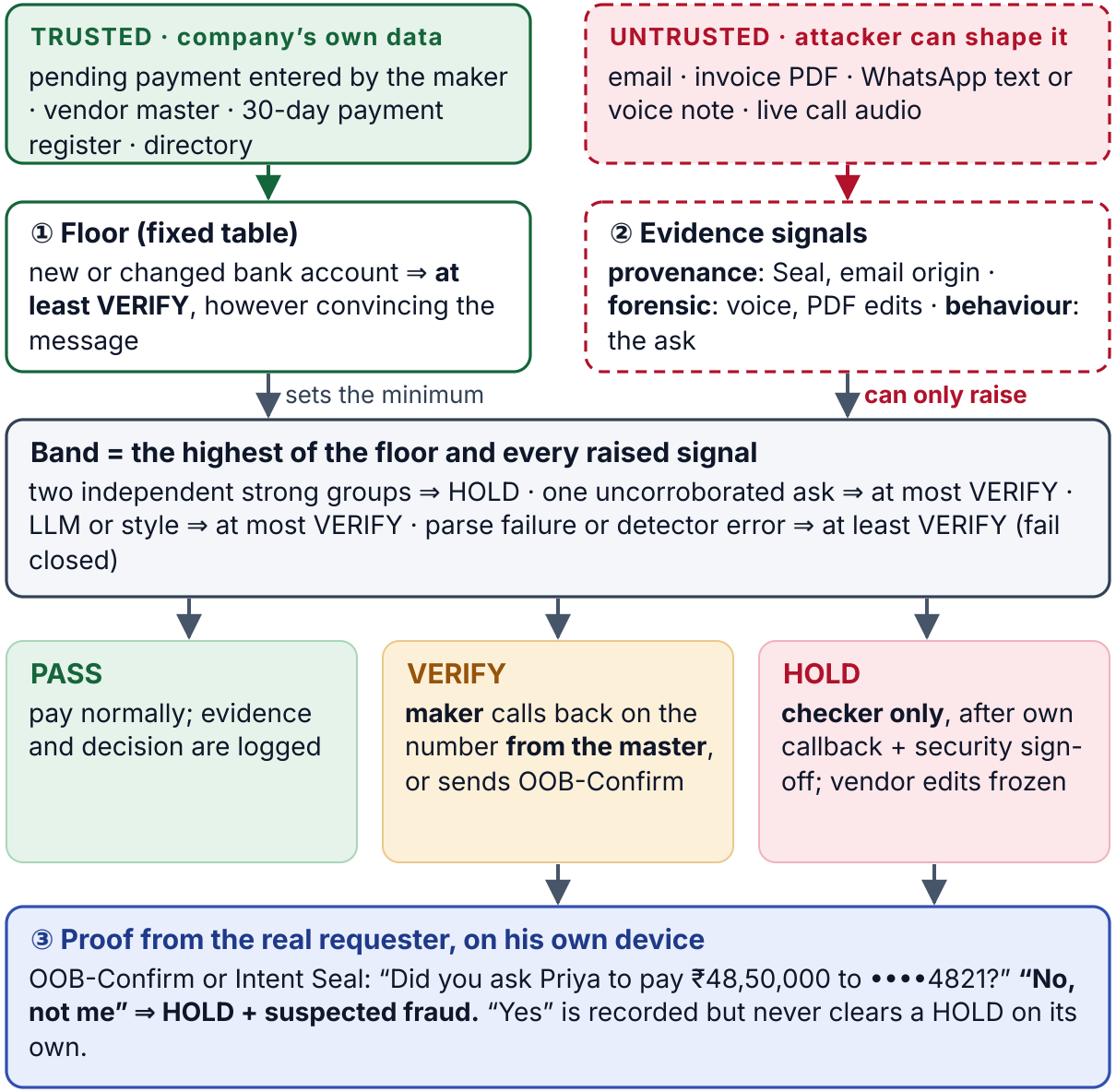
USER VALUE

Priya **never has to judge whether a voice is fake**. Routine known-vendor payments add no step unless evidence raises them: 3.0% [1.4-6.4] of routine payments reached the checker in our synthetic test (n = 200, X1).

OTHERS DO	WHAT THAT GIVES YOU	WHAT SATYAPAN DOES DIFFERENTLY
Passive deepfake detectors (meeting, contact-centre and upload tools)	a probability that decays: on 2026 video generators the best detector scored 69.7% AUROC [R10]	**Proves the real executive** (Seal); no score is ever the gate
Meeting identity badges (Zoom + World ID beta, RealityCheck) [R16]	verify the *account*, on one platform; not the audio, not the payment	seals the **voice itself** to a passkey, this meeting, this 5 s; works for voice notes
In-band caller authentication research (CallShield, 2026) [R13]	about 63 s to authenticate; keys shared between contacts	5 s median verdict offline (X1); delayed key disclosure: receivers cannot forge
Callback and maker-checker (standard BEC hygiene)	right idea, done by hand, often skipped; number taken from the message	enforced and logged; number only from the master; **moved to the checker** when anything looks off

[R2] SEBI PR 40/2026 · [R10] DF26, arXiv 2609.07369 · [R13] CallShield, arXiv 2601.09327 · [R16] Zoom + World ID (May 2026), Beyond Identity RealityCheck (Aug 2024)

A trusted floor, raise-only evidence, and humans the attacker has not reached



Policy engine: `src/satyapan/policy/` · invariants: `tests/invariants/` · full signal → band table and c

FLOOR TABLE IN PLAIN WORDS (TRUSTED INPUTS ONLY)

IF THE PAYMENT GOES TO...	FLOOR	BEFORE PAYING
a known vendor, same account, usual amount	PASS	nothing extra
a known vendor, unusual amount or dormant 180+ days	VERIFY	callback or OOB-Confirm
a new payee, up to ₹25,000 incl. last 30 days	PASS	bank's own new-payee checks
a new payee, above ₹25,000	VERIFY	callback or OOB; above ₹10 lakh both
a vendor whose bank account changed	VERIFY	callback on the **old** number; bank-shown name = master
(a change to a phone, email or approver on file)	HOLD	checker on the old channel + 2nd approver

SAFETY RULES, PROPERTY-TESTED ON THE REAL ENGINE (7 TEST FILES)

- Untrusted content can only raise the band, never lower it.
- The floor reads only trusted data (fuzzed with hostile inputs).
- LLM or style signals: at most VERIFY, never counted as strong.
- No HOLD from a single uncorroborated ask (policy 1.1).
- Only the checker can clear a HOLD; the maker never can.
- No code path files a report; a human always submits.

Detectors decide who verifies, never whether money to a new or changed account needs verifying.

A deepfake works by manipulating the person who would verify. Any strong signal therefore moves the decision to someone else